

Device Fingerprinting — Day1 & Day2

Hakim Lado 2026-06-13

Objective

Identify which transmitter sent a captured signal from the raw I/Q waveform.

Two datasets, evaluated independently:

Day1: B200_1, B200_2, B205_1, B205_2 Day2: 2922, B200, B205, ettus

Each device is fingerprinted in two representations (time, frequency) x two model families (real 1D-CNN, complex-valued CVNN) – four configurations per dataset.

Method (brief)

- complex64 I/Q at $F_s = 5$ MHz, 4 devices x 3 runs (~35 s each).
- Bursts energy-detected, sliced into 512-sample power-normalized windows.
- Leakage-safe split: train/val on runs 1–2, test on the unseen run 3.
- Frequency representation = FFT of each window.
- Window-level accuracy (per slice) and frame-level accuracy (vote per burst).

Findings

The frequency-domain 1D-CNN is the best model on both datasets (99.3% Day1, 95.4% Day2, frame-level). The frequency representation beat the time domain in every configuration. Day2 is harder; its main error is 'ettus' vs '2922'.

Table 1. Day1 test accuracy (run 3 held out)

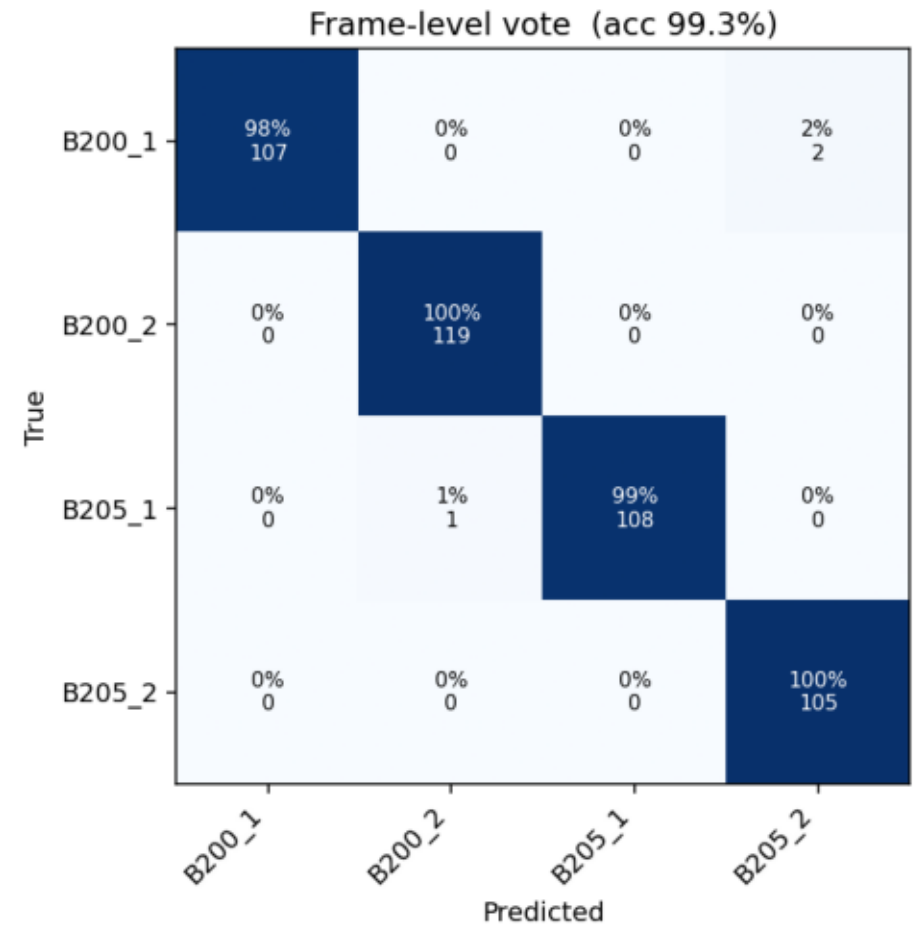
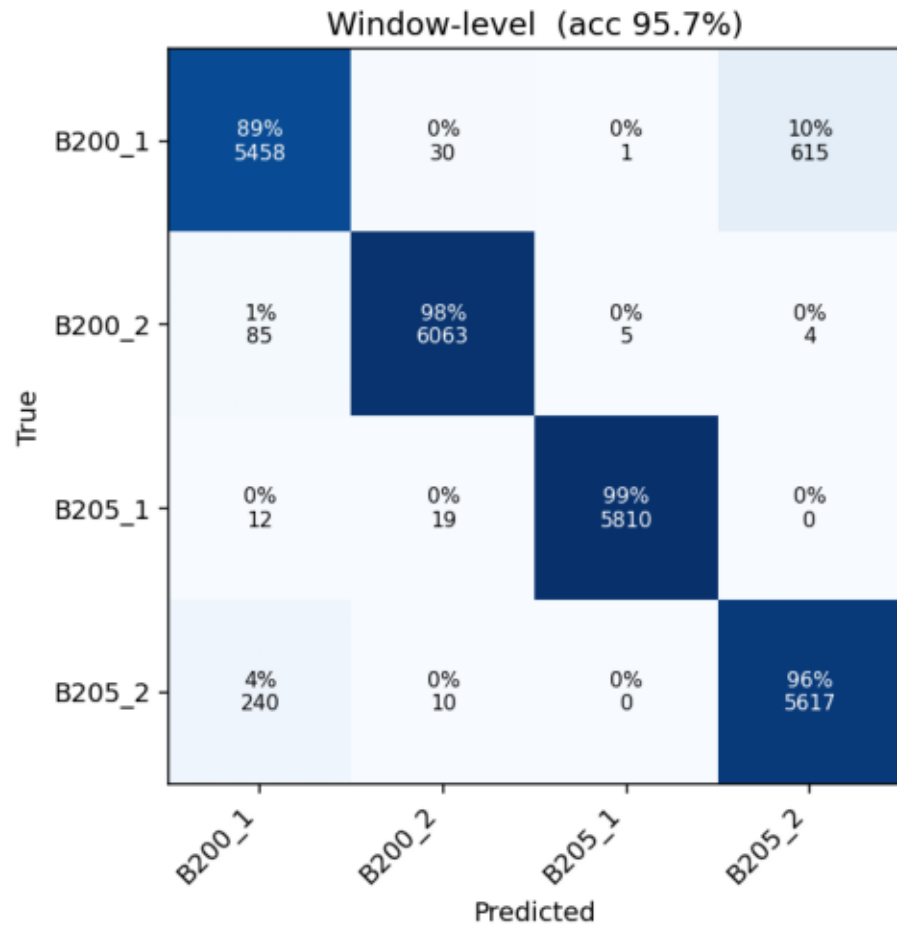
Configuration	Window	Frame
Time-domain · 1D-CNN	93.8%	97.1%
Time-domain · CVNN	89.9%	94.3%
Freq-domain · 1D-CNN	95.7%	99.3%
Freq-domain · CVNN	94.9%	98.6%

Table 2. Day2 test accuracy (run 3 held out)

Configuration	Window	Frame
Time-domain · 1D-CNN	84.9%	92.7%
Time-domain · CVNN	83.3%	90.0%
Freq-domain · 1D-CNN	91.9%	95.4%
Freq-domain · CVNN	83.2%	93.3%

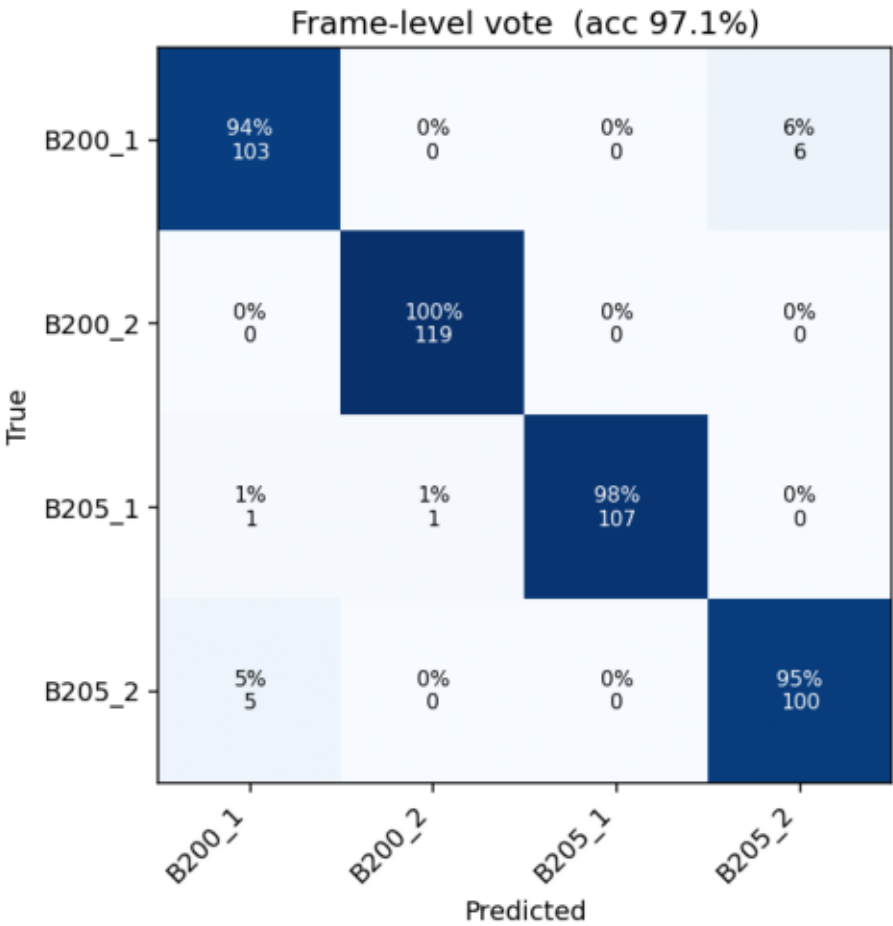
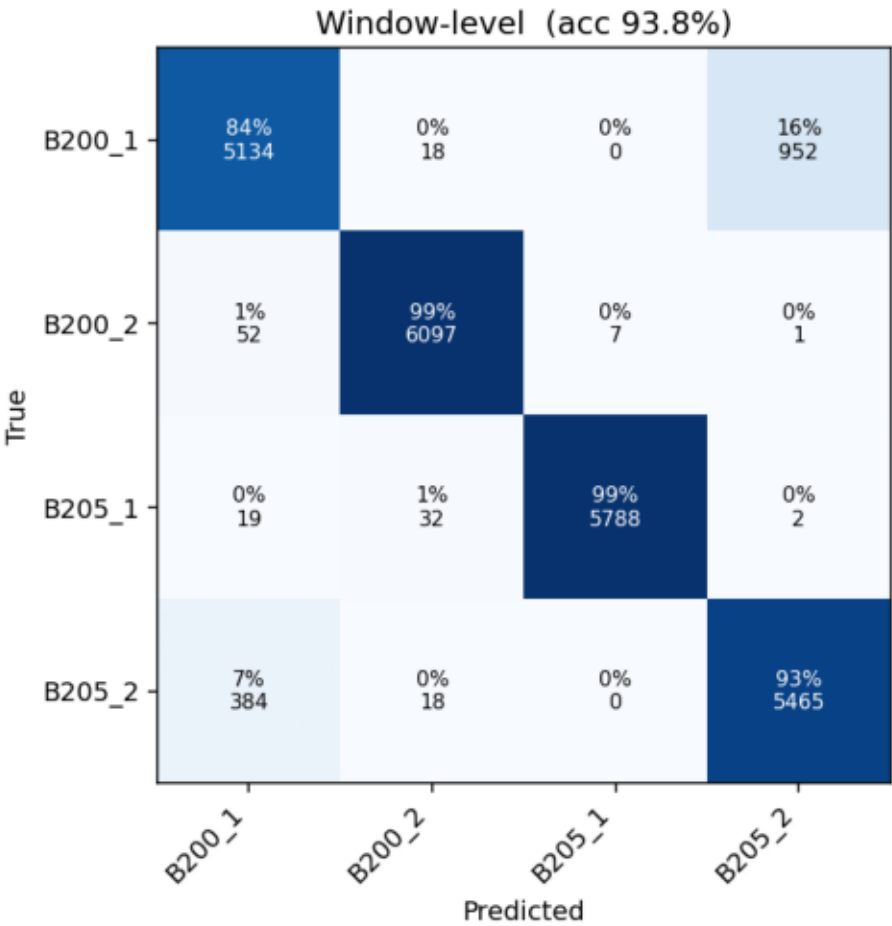
Day1 — Confusion Matrix — Freq-domain 1D-CNN (best)

Confusion matrix — freq_cnn1d



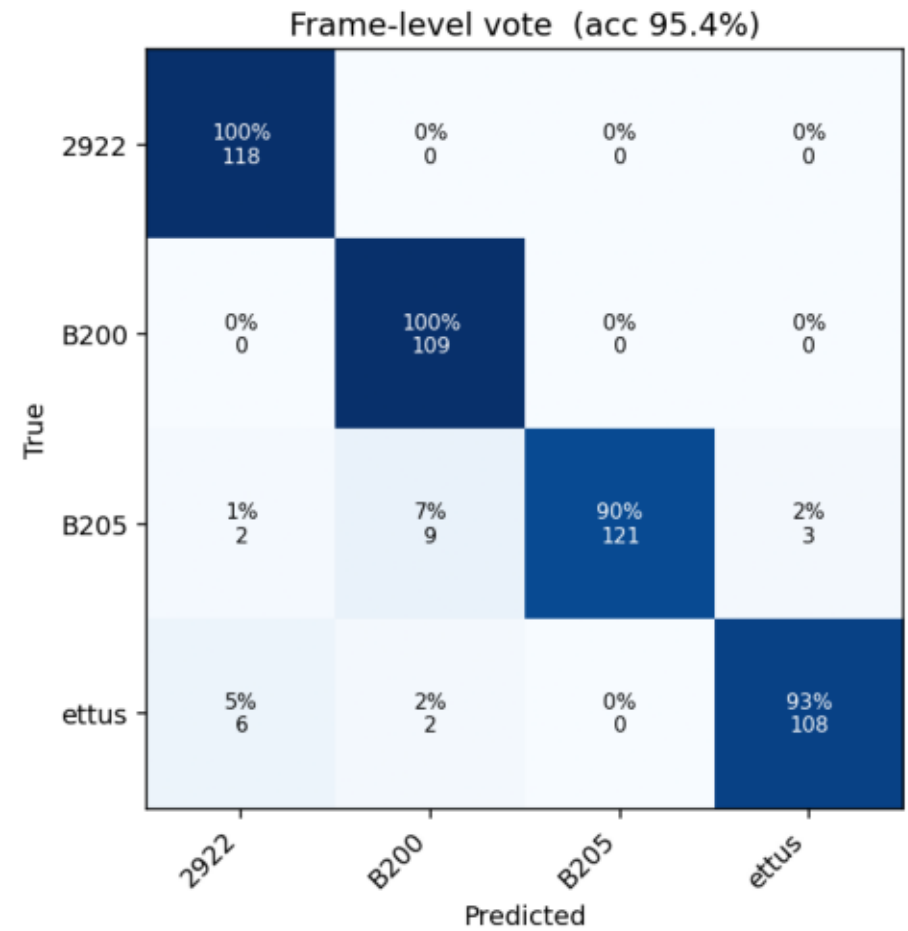
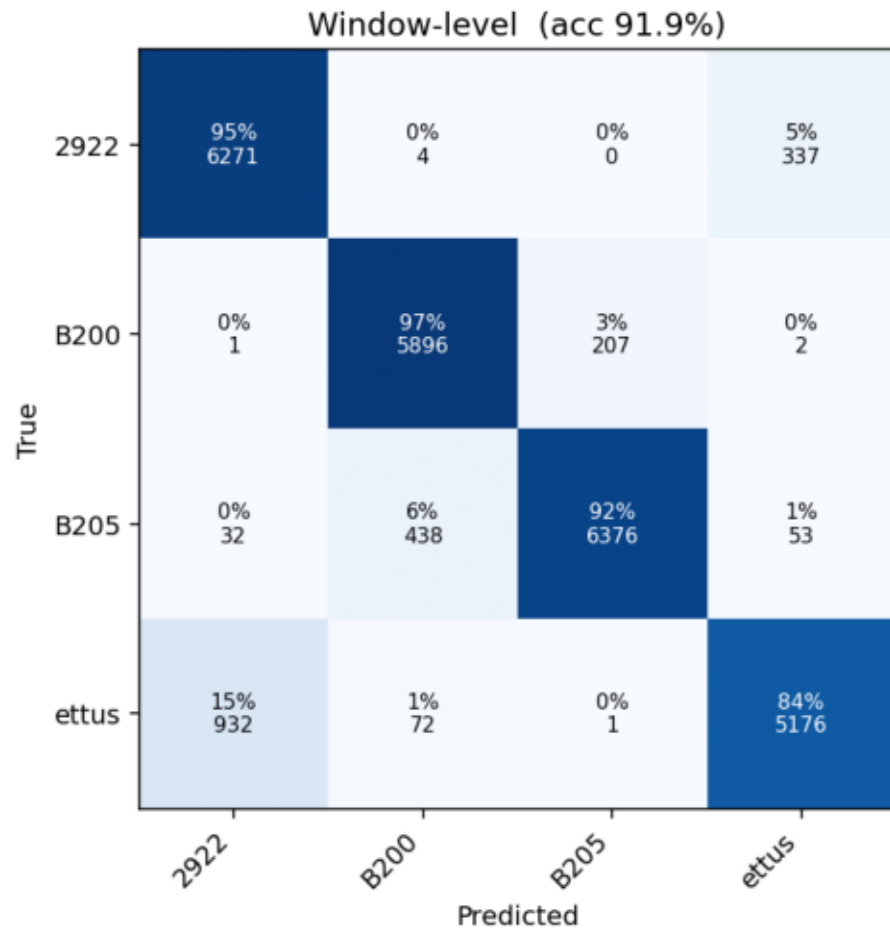
Day1 — Confusion Matrix — Time-domain 1D-CNN

Confusion matrix — time_cnn1d



Day2 — Confusion Matrix — Freq-domain 1D-CNN (best)

Confusion matrix — freq_cnn1d



Day2 — Confusion Matrix — Time-domain 1D-CNN

Confusion matrix — time_cnn1d

